

FA-KGD: Frequency-Aware Kalman-Guided Diffusion for Accelerated MRI Reconstruction



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Introduction & Motivation

Accelerated MRI reduces scan time by acquiring only a fraction of k-space, making reconstruction a severely underdetermined inverse problem. **Diffusion priors** are attractive because one pretrained model can be reused across anatomies, masks, and acceleration factors without retraining.

However, training-free diffusion samplers suffer from three MRI-specific limitations:

- **Isotropic noise assumption:** all k-space measurements treated as equally reliable.
- **Data-consistency (DC) shock:** high frequencies enforced too early during denoising.
- **Open-loop calibration:** noise estimates fixed before inference, unable to adapt.

Our solution: FA-KGD — one training-free component per limitation: frequency-adaptive Kalman gain, frequency-progressive DC, and on-line noise refinement. No retraining, no learned parameters, <1% overhead.

Dataset & Setup

fastMRI benchmark

- **Brain:** 192 AXFLAIR slices, 12 validation volumes (domain-matched prior)
- **Knee:** 192 single-coil slices, 6 volumes (cross-domain, brain-trained prior)

Protocol

- Variable-density Cartesian undersampling, $R \in \{4, 8\}$
- Fully sampled 24×24 ACS region
- Volume-wise normalization; PSNR (dB) and SSIM

Baselines: DPS, ADPS, IIGDM (primary); supervised reference: E2E-VarNet.

Research questions

1. Does FA-KGD improve reconstruction quality?
2. Does performance scale with diffusion steps?
3. Does it generalize across anatomies?

Methods

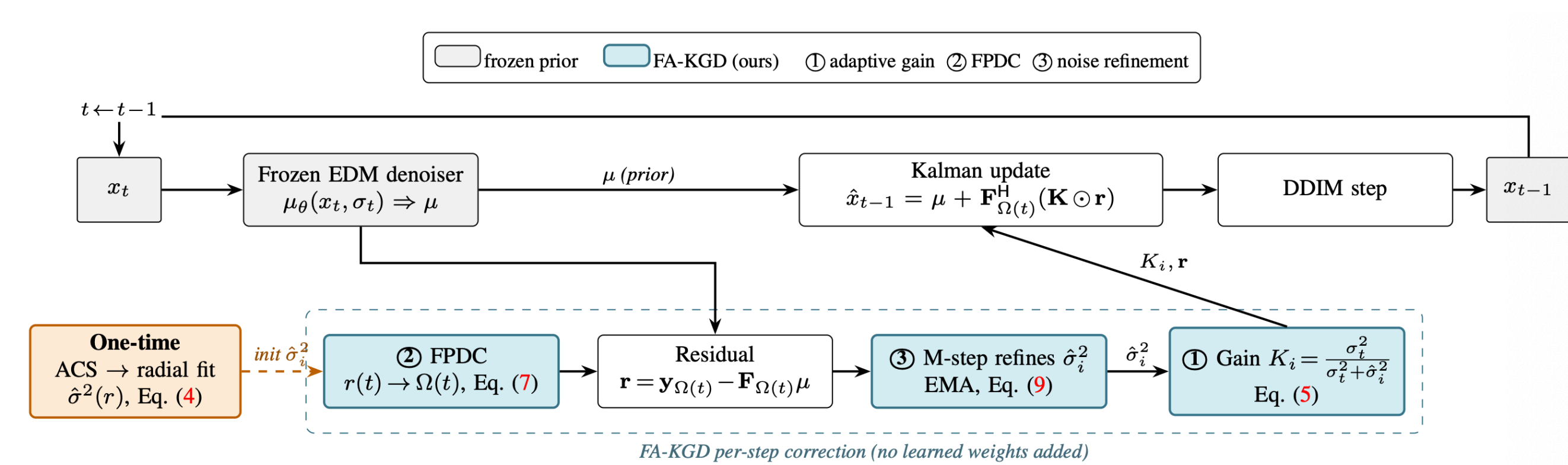


Figure 1. FA-KGD per-step pipeline: a frozen EDM denoiser produces the prior estimate μ ; the three FA-KGD components form a frequency-adaptive, progressively gated Kalman correction before the DDIM step.

1. Frequency-Adaptive Kalman Gain. A *one-time* calibration fits a radial noise model $\hat{\sigma}^2(r; \theta) = \theta_0 + \sum_k \theta_k (r/r_{\max})^{2k}$ to ACS replicate variances (3 parameters per volume, milliseconds). Replacing the scalar σ_y^2 with the map $\hat{\sigma}_i^2$ gives a per-frequency gain and update

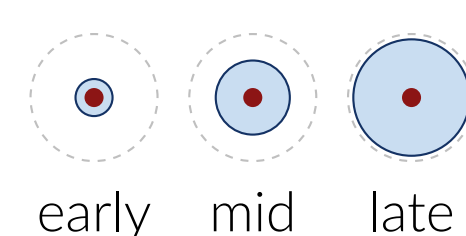
$$K_i(t) = \frac{\sigma_t^2}{\sigma_t^2 + \hat{\sigma}_i^2}, \quad \hat{x}_{t-1} = \mu + \mathbf{F}_{\Omega(t)}^H (\mathbf{K}(t) \odot \mathbf{r}).$$

Early steps and clean frequencies: $K_i \rightarrow 1$ (trust data); late steps and noisy frequencies: $K_i \rightarrow 0$ (trust prior).

2. Frequency-Progressive Data Consistency (FPDC). Hard DC at all frequencies forces residuals to zero that the denoiser expects to be noisy (“DC shock”). FPDC enforces DC only inside an expanding radius

$$r(t) = r_{\text{ACS}} + (r_{\max} - r_{\text{ACS}}) \left(1 - \frac{t}{T}\right)^\beta,$$

matching the coarse-to-fine order in which diffusion recovers structure: low frequencies first, high frequencies later.



Blue disk: active frequencies $\Omega(t)$. Red core: ACS, always on. Dashed: $r_{\max}$.

3. Online Noise Refinement. The noise map is refined during sampling from the measurement residual via a time-decaying EMA (M-step):

$$\hat{\sigma}_i^2 \leftarrow \alpha_t \hat{\sigma}_i^2 + (1 - \alpha_t) \max(|r_i|^2 - \gamma \sigma_t^2, \epsilon).$$

$\gamma=0$ is the safe default: full bias correction ($\gamma=1$) drives $\hat{\sigma}_i^2 \rightarrow 0$, collapsing the gain (−12.8 dB in simulation).

Choice	Prior samplers	FA-KGD
Measurement noise	scalar σ_y^2	frequency map $\hat{\sigma}_i^2$
DC mask	fixed Ω	progressive $\Omega(t)$
Sampling gain	fixed / open loop	residual-refined
Learned parameters	none added	none added

Table 1. How FA-KGD modifies the posterior-sampling update.

FA-KGD-3D. Pooling ACS statistics across the N_z slices of a volume multiplies the effective degrees of freedom by N_z , cutting noise-map RMSE from 26% to 6.3% at zero extra cost. Defaults throughout: $\alpha_0=0.95$, $\beta=1$, $\gamma=0$.

Results

Setting	T	DPS	ADPS	IIGDM	FA-KGD
Brain $R=4$	20	26.64	25.26	29.83	30.06
Brain $R=4$	50	26.90	25.34	29.85	30.24
Brain $R=8$	20	24.93	22.74	27.14	27.24
Brain $R=8$	50	25.25	23.09	27.13	27.33
Knee $R=4$	20	29.29	27.18	31.27	31.51
Knee $R=8$	20	26.49	24.18	28.33	28.47

Table 2. PSNR (dB, $\uparrow$) on fastMRI: brain (domain-matched) and knee (cross-domain with the brain-trained prior). Best training-free method in bold — FA-KGD wins in every setting.

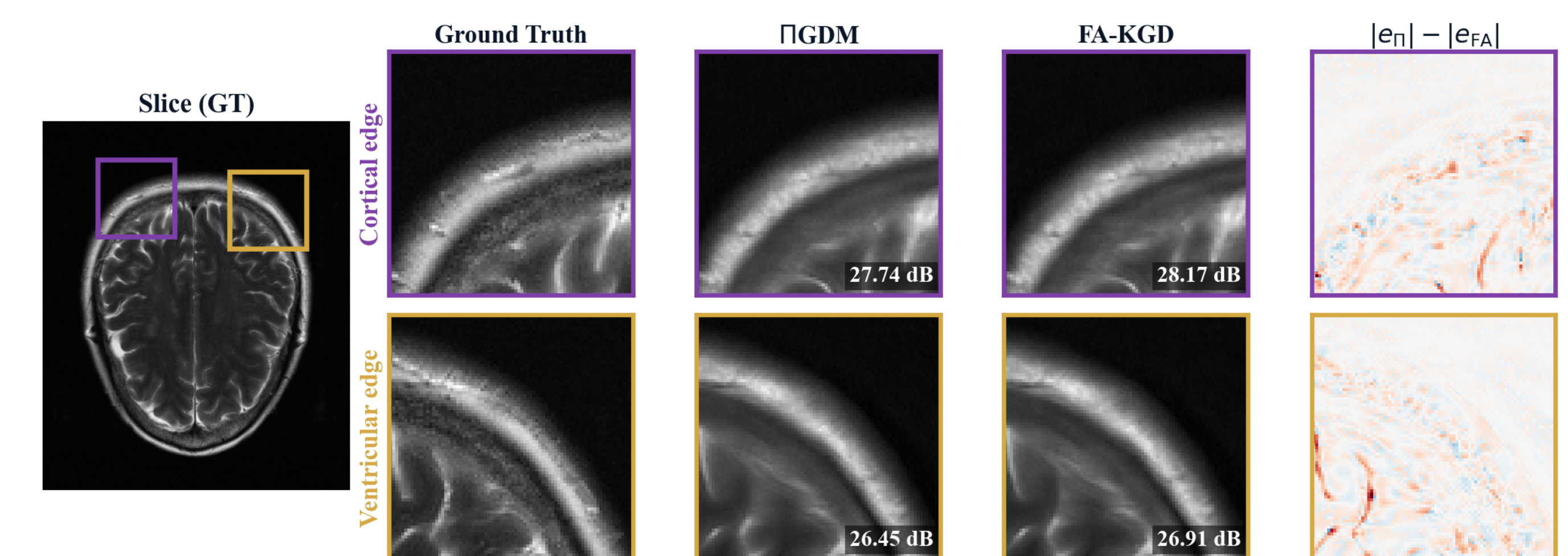


Figure 2. Two highest-variance edge crops (cortical, ventricular) of a brain $R=4$, $T=50$ slice: ground truth, IIGDM, FA-KGD, and the signed error map (red = FA-KGD better). FA-KGD improves localised PSNR at both edge crops by roughly +0.4 to +0.5 dB over IIGDM.

Analysis

Why does FA-KGD work? The adaptive gain accounts for frequency-dependent uncertainty, FPDC removes early high-frequency over-correction, and online refinement keeps the calibration consistent throughout sampling.

Scaling with the step budget T (brain $R=4$):

T	10	20	50	100
Δ PSNR vs. IIGDM (dB)	+0.11	+0.23	+0.39	+0.51

IIGDM peaks at $T=50$ and then degrades — the signature of accumulated DC shock — while FA-KGD improves monotonically and the gap widens strictly with T . The same pattern replicates cross-domain on knee.

Conclusions

- Three plug-in, training-free components fix MRI-specific failure modes of diffusion samplers; no retraining, no added parameters, <1% overhead.
- Best training-free method in every setting; gains grow with the diffusion budget and are visible at clinically relevant edges.
- **Future work:** multi-coil reconstruction, learned covariance models, clinical validation, other inverse problems.